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## Kinematics 7: $v^2 - u^2 = 2as$

One equation that links speeds and distance, with no time in it at all.

### Background Knowledge

Read this as you come in. Highlight the key words, and use the space to add your own notes.

#### A. Distance, speed and velocity

|                 | Scalar — size only                        | Vector — size <b>and</b> direction                          |
|-----------------|-------------------------------------------|-------------------------------------------------------------|
| <b>How far</b>  | <b>distance</b> , in metres (m)           | <b>displacement</b> : how far from the start, and which way |
| <b>How fast</b> | <b>speed</b> , in metres per second (m/s) | <b>velocity</b> : speed in a stated direction               |

$$s = vt \quad \text{distance travelled} = \text{speed} \times \text{time}$$

A car at 20 m/s for 5.0 s travels  $20 \times 5.0 = 100$  m. This works when the speed is **steady**. When the speed changes, use the **average** speed.

#### B. Acceleration

**Acceleration** is the **rate of change of velocity**: how much the velocity changes each second.

$$a = \frac{\Delta v}{t}$$

acceleration = change in velocity  $\div$  time

$$\Delta v = v - u \quad (\text{final} - \text{initial})$$

- $u$  is the **initial** velocity,  $v$  the **final** velocity. Unit of acceleration: **m/s<sup>2</sup>** (metres per second, every second).
- Speeding up:  $a$  is **positive**. Slowing down:  $a$  is **negative**. Finish minus start, in that order, and the sign comes out right by itself.
- **Uniform** acceleration means the velocity changes by the same amount every second. An object falling freely near the Earth accelerates at about **9.8 m/s<sup>2</sup>**.

A bike goes from 2.0 m/s to 8.0 m/s in 3.0 s:  $a = (8.0 - 2.0) \div 3.0 = 2.0 \text{ m/s}^2$ .

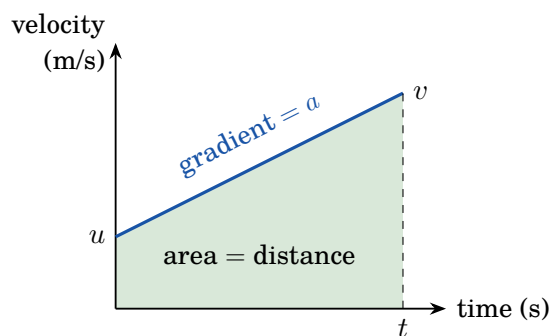
### C. Reading the motion graphs

#### Distance–time graph

- **gradient = speed**
- a flat line: stationary
- a steeper line: faster

#### Velocity–time graph

- **gradient = acceleration**
- **area under the line = distance travelled**
- a flat line: constant velocity



$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

A straight sloping line on a velocity–time graph means **uniform** acceleration. That is the case today's equation is built for.

### D. How this ties together — and where today goes

Everything so far has **time** in it:  $s = vt$ , and  $a = \Delta v \div t$ .

Plenty of real questions give you **no time at all**. A car brakes from 20 m/s to a stop: how far does it skid? An aeroplane needs to reach 60 m/s: how long must the runway be? You know the speeds and the acceleration, and you want the distance.

Two equations you already have, written with  $u$  and  $v$ :

①  $s = \frac{1}{2}(u + v)t$       average velocity  $\times$  time: the area under the graph in C

②  $v = u + at$       acceleration from B, rearranged

Both have  $t$  in them. Put them together and the  $t$  drops out, leaving one equation with no time in it:

$$v^2 - u^2 = 2as$$

**Today's question:** if you know how fast something starts, how fast it finishes and how hard it accelerates, how far does it travel?

### The equation — it is on your equation sheet

$$v^2 - u^2 = 2 a s$$

(final velocity)<sup>2</sup> – (initial velocity)<sup>2</sup> = 2 × acceleration × distance

| Symbol | Quantity         | Unit             |
|--------|------------------|------------------|
| $s$    | distance         | m                |
| $u$    | initial velocity | m/s              |
| $v$    | final velocity   | m/s              |
| $a$    | acceleration     | m/s <sup>2</sup> |
| $t$    | time             | s                |

### The Givens — start every calculation with this box

#### Givens

$s =$   
 $u =$   
 $v =$   
 $a =$   
 $t =$

Write the five letters **down the page**, always in this order:  $s, u, v, a, t$ . Read downwards, they spell **SUVAT**.

- Fill in each one the question gives you, with its unit.
- Put ? next to the one you want.
- Leave the others empty.

The **empty** line is the quantity the question leaves out. When  $t$  is empty, use  $v^2 - u^2 = 2 a s$ : the equation with **no**  $t$ .

### Worked example 1 — find the distance

A cyclist speeds up from 3.0 m/s to 7.0 m/s with an acceleration of 0.50 m/s<sup>2</sup>. Calculate the distance travelled while accelerating.

#### 1. Givens

$s = ?$   
 $u = 3.0 \text{ m/s}$   
 $v = 7.0 \text{ m/s}$   
 $a = 0.50 \text{ m/s}^2$   
 $t =$

**2. Equation**  $v^2 - u^2 = 2 a s$

**3. Substitute**  $7.0^2 - 3.0^2 = 2 \times 0.50 \times s$

**4. Rearrange**  $s = \frac{7.0^2 - 3.0^2}{2 \times 0.50}$

**5. Calculate**  $s = \frac{49 - 9.0}{1.0} = \mathbf{40 \text{ m}}$

### Worked example 2 — find the final velocity

A trolley moving at 5.0 m/s accelerates at 2.0 m/s<sup>2</sup> over a distance of 30 m. Calculate its final velocity. Give your answer to 2 significant figures.

#### 1. Givens

$s = 30 \text{ m}$   
 $u = 5.0 \text{ m/s}$   
 $v = ?$   
 $a = 2.0 \text{ m/s}^2$   
 $t =$

**2. Equation**  $v^2 - u^2 = 2 a s$

**3. Substitute**  $v^2 - 5.0^2 = 2 \times 2.0 \times 30$

**4. Rearrange**  $v^2 = 5.0^2 + 2 \times 2.0 \times 30$   
 $v = \sqrt{5.0^2 + 2 \times 2.0 \times 30}$

**5. Calculate**  $v = \sqrt{25 + 120} = \sqrt{145}$   
 $= 12.04 \dots = \mathbf{12 \text{ m/s}} \text{ (2 s.f.)}$

**The check.** Put your answer back into  $v^2 - u^2 = 2 a s$  and work out **each side on its own**. The two sides give the **same number**. In worked example 1:  $v^2 - u^2 = 49 - 9 = 40$  and  $2 a s = 2 \times 0.50 \times 40 = 40$ . They match, so  $s = 40 \text{ m}$  is right.

## Questions

Fill in the Givens box first, then follow the worked examples: equation, substitute under each letter, rearrange, calculate, and give the answer with its unit. Use the check on at least one answer in each section.

### Everyone — find the distance

1) **Calculate** the distance a runner covers while accelerating from rest to 8.0 m/s at 2.0 m/s<sup>2</sup>. [3]

#### Givens

$$s = ?$$

$$u = 0$$

$$v = 8.0 \text{ m/s}$$

$$a = 2.0 \text{ m/s}^2$$

$$t =$$

$$v^2 - u^2 = 2 a s$$

$$8.0^2 - 0^2 = 2 \times 2.0 \times s$$

$$s = \frac{8.0^2 - 0^2}{2 \times 2.0} = \frac{64}{4.0} = \mathbf{16 \text{ m}}$$

2) **Calculate** the distance a car travels while it speeds up from 6.0 m/s to 14 m/s with an acceleration of 1.6 m/s<sup>2</sup>. [3]

#### Givens

$$s = ?$$

$$u = 6.0 \text{ m/s}$$

$$v = 14 \text{ m/s}$$

$$a = 1.6 \text{ m/s}^2$$

$$t =$$

$$v^2 - u^2 = 2 a s$$

$$14^2 - 6.0^2 = 2 \times 1.6 \times s$$

$$s = \frac{14^2 - 6.0^2}{2 \times 1.6} = \frac{160}{3.2} = \mathbf{50 \text{ m}}$$

3) **Calculate** the distance a skateboarder rolls downhill while speeding up from 1.0 m/s to 5.0 m/s with an acceleration of 0.80 m/s<sup>2</sup>. [3]

#### Givens

$$s = ?$$

$$u = 1.0 \text{ m/s}$$

$$v = 5.0 \text{ m/s}$$

$$a = 0.80 \text{ m/s}^2$$

$$t =$$

$$v^2 - u^2 = 2 a s$$

$$5.0^2 - 1.0^2 = 2 \times 0.80 \times s$$

$$s = \frac{5.0^2 - 1.0^2}{2 \times 0.80} = \frac{24}{1.6} = \mathbf{15 \text{ m}}$$

**4) Calculate** the braking distance of a car that slows from 20 m/s to a stop with an acceleration of  $-5.0 \text{ m/s}^2$ . A distance comes out positive. [3]

**Givens**

$$s = ?$$

$$u = 20 \text{ m/s}$$

$$v = 0$$

$$a = -5.0 \text{ m/s}^2$$

$$t =$$

$$v^2 - u^2 = 2 a s$$

$$0^2 - 20^2 = 2 \times (-5.0) \times s$$

$$s = \frac{0^2 - 20^2}{2 \times (-5.0)} = \frac{-400}{-10} = 40 \text{ m}$$

**5) Calculate** the acceleration of an aeroplane that starts from rest and reaches 60 m/s along a runway 900 m long. [3]

**Givens**

$$s = 900 \text{ m}$$

$$u = 0$$

$$v = 60 \text{ m/s}$$

$$a = ?$$

$$t =$$

$$v^2 - u^2 = 2 a s$$

$$60^2 - 0^2 = 2 \times a \times 900$$

$$a = \frac{60^2 - 0^2}{2 \times 900} = \frac{3600}{1800} = 2.0 \text{ m/s}^2$$

**6) Calculate** the final velocity of a ball that is rolling at 3.0 m/s and then accelerates at  $1.5 \text{ m/s}^2$  over 12 m. Give your answer to 2 significant figures. [3]

**Givens**

$$s = 12 \text{ m}$$

$$u = 3.0 \text{ m/s}$$

$$v = ?$$

$$a = 1.5 \text{ m/s}^2$$

$$t =$$

$$v^2 - u^2 = 2 a s$$

$$v^2 - 3.0^2 = 2 \times 1.5 \times 12$$

$$v = \sqrt{3.0^2 + 2 \times 1.5 \times 12} = \sqrt{45}$$

$$= 6.708 \dots = 6.7 \text{ m/s}$$

**Stretch** — two equations in one question

7) A stone is dropped from rest from a bridge and falls 20 m to the water. It falls with an acceleration of  $9.8 \text{ m/s}^2$ .

- (a) Calculate the speed of the stone as it reaches the water. Give your answer to 2 significant figures.

**Givens**

$$s = 20 \text{ m}$$

$$u = 0$$

$$v = ? \text{ (a)}$$

$$a = 9.8 \text{ m/s}^2$$

$$t = ? \text{ (b)}$$

$$v^2 - u^2 = 2 a s$$

$$v^2 - 0^2 = 2 \times 9.8 \times 20$$

$$v = \sqrt{0^2 + 2 \times 9.8 \times 20} = \sqrt{392}$$

$$= 19.80 \dots = \mathbf{20 \text{ m/s}} \text{ (2 s.f.)}$$

[3]

- (b) Use your answer to (a) and  $a = \frac{\Delta v}{t}$  to calculate how long the stone takes to fall.

$$t = \frac{\Delta v}{a} = \frac{19.8 - 0}{9.8} = 2.02 \dots = \mathbf{2.0 \text{ s}}$$

[2]

8) A train speeds up steadily from  $8.0 \text{ m/s}$  to  $16 \text{ m/s}$  over a distance of  $240 \text{ m}$ .

- (a) Calculate the acceleration of the train.

**Givens**

$$s = 240 \text{ m}$$

$$u = 8.0 \text{ m/s}$$

$$v = 16 \text{ m/s}$$

$$a = ? \text{ (a)}$$

$$t = ? \text{ (b)}$$

$$v^2 - u^2 = 2 a s$$

$$16^2 - 8.0^2 = 2 \times a \times 240$$

$$a = \frac{16^2 - 8.0^2}{2 \times 240} = \frac{192}{480} = \mathbf{0.40 \text{ m/s}^2}$$

[3]

- (b) Calculate the time the train takes to cover the  $240 \text{ m}$ .

$$t = \frac{\Delta v}{a} = \frac{16 - 8.0}{0.40} = \mathbf{20 \text{ s}}$$

[2]

- (c) Calculate the average speed of the train over the  $240 \text{ m}$ . Your answer sits between its initial and final velocities.

$$\frac{240}{20} = \mathbf{12 \text{ m/s}} \text{ — between } 8.0 \text{ and } 16, \text{ and exactly halfway, as it is for uniform acceleration.}$$

[1]